

JEE MAIN REVISION MASTER: GRAVITATION 2026

2 QUESTIONS 3.33% WEIGHTAGE

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 14 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Kcet Gravitation - Core Application	14 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Gravitation - Core Application (High Yield — 14 PYQs)

Core Principle: Key problem pattern in Kcet Gravitation.

■■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "Suppose the acceleration due to gravity at the earth's surface is gm/s^2 and at the surface of moon it is g^m/s^2 " [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "Imagine a new planet having the same density as that of the earth, but it is two times bigger than the earth in size. If the acceleration du..." [Ans: N/A]
- [2025 Shift] Q#3 (MCQ): "If $r_{(p)}, v_{(p)}, L_{(p)}$ and $r_{(a)}, v_{(a)}, L_{(a)}$ are radii, velocities and angular momenta of a planet at perihelion and aphelion of its ellip..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

Suppose the acceleration due to gravity at the earth's surface is $g\text{ m/s}^2$ and at the surface of moon it is $g'\text{ m/s}^2$. An M kg passenger goes from the earth to the moon in a spaceship moving with a constant velocity (Neglect all other objects in the sky). Which curve best represents the weight (net gravitational force) as a function of time?

- (A)
- (B)
- (C)
- (D)

Correct Answer: None

Solution Analysis: Let the mass of the Earth be M_e , the mass of the Moon be M_m , and the distance between their centres be d . Suppose the spaceship is at a distance r from the centre of the Earth. Then its distance from the centre of the Moon is $(d-r)$. The Earth pulls the passenger towards the Earth, and the Moon pulls the passenger towards the Moon. These two pulls are in opposite directions, so the net gravitational force on the passenger of mass M is:
$$F_{\text{net}} = \frac{GM_e M}{r^2} - \frac{GM_m M}{(d-r)^2}$$
 The spaceship moves with constant velocity. So the distance r from the Earth in